



8/13/2026

ASSESSMENT 1

CompTIA N+

N+ Network Security, Maintenance &
Troubleshooting Procedures.

⌘ Section A – Concept Application

Scenario 1.

❖ **Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.**

❖ **What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?**

➤ **CPU, RAM and Storage Selection**

For employees who use **programming software, web browsers, and video conferencing at the same time**, the following factors should be considered:

- ⌘ **CPU:** Choose a modern multi-core processor such as **Intel Core i5/i7 or AMD Ryzen 5/7** for smooth multitasking.
- ⌘ **RAM:** At least 16 GB RAM is recommended so multiple applications can run without slowing down the computer.
- ⌘ **Storage:** A **512 GB or larger SSD** should be preferred because it provides faster boot and application loading compared to an HDD.
- ⌘ **Compatibility:** The CPU, RAM, and storage must be compatible with the **motherboard**, including the supported processor socket, RAM type, and storage interface.

Conclusion: A modern multi-core CPU, **16 GB RAM, and an SSD** would provide good performance and reliable hardware compatibility for the employees' workload.

Scenario 2.

❖ After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

❖ Question: Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

➤ **Antivirus and System Performance**

Antivirus software continuously runs in the background to **scan files, programs, and system activity**. This can affect system resources in the following ways:

- ⌘ **CPU:** Antivirus scans use CPU processing power, leaving less CPU capacity for other applications and making the system slower.
- ⌘ **RAM:** The antivirus keeps its services and security processes in memory, increasing RAM usage and reducing memory available for other programs.
- ⌘ **Storage:** Antivirus software frequently reads and scans files on the storage drive. This increases **disk usage** and can slow down file access and application loading.

Conclusion: Antivirus software improves security, but its background scanning can consume **CPU, RAM, and storage resources**, which may cause noticeable system slowdowns.

Scenario 3.

❖ **A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.**

❖ **Question: Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.**

➤ **Driver and OS Troubleshooting:**

- **Check the USB connection:** Connect the keyboard and printer properly and try different USB ports.
- **Check Device Manager:** Look for any warning marks or errors related to the devices.
- **Update drivers:** Download and install the correct, compatible drivers for the keyboard and printer.
- **Reinstall drivers:** If updating does not work, uninstall the faulty driver and reinstall it.
- **Check the operating system:** Make sure the OS is updated and supports the connected devices.
- **Restart the computer:** Restart the system after installing or updating the drivers.

Conclusion: The problem can usually be fixed by installing **compatible drivers**, updating the operating system, and restarting the computer.

Scenario 4.

- ❖ Nexus Innovations experiences frequent power cuts and voltage fluctuations.
- ❖ Question: Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

➤ **UPS and Surge Protector:**

- ⌘ **UPS (Uninterruptible Power Supply):** A UPS provides temporary backup power when electricity goes out. It gives employees enough time to **save their work and shut down computers safely**, preventing data loss and sudden hardware damage.
- ⌘ **Surge Protector:** A surge protector protects computer components from sudden voltage spikes or fluctuations by limiting excess voltage before it reaches the hardware.

Conclusion: Using both a **UPS and surge protector** helps protect computer hardware from power problems and reduces the risk of **data loss and system damage**.

Scenario 5.

- ❖ New employees at Nexus Innovations report that they can reach internal systems (e.g. the application server) by IP address but not by name, and their devices sometimes fail to obtain a valid IP address when they connect to the network.
- ❖ Question: Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

➤ DNS and DHCP

How DNS and DHCP Work Together:

- ⌘ **DHCP** automatically provides devices with an **IP address, subnet mask, default gateway, and DNS server address** when they connect to the network.
- ⌘ **DNS** converts a system name, such as applicationserver.local, into its corresponding **IP address**, allowing users to access the server by name.

Most Likely Misconfigured Service

The symptoms point to two possible issues:

- ⌘ **Cannot access by name but can access by IP:** This indicates a **DNS problem**.
- ⌘ **Devices sometimes fail to obtain a valid IP address:** This indicates a **DHCP problem**.

Conclusion: Both services may need checking, but the **DNS service is responsible for the name-resolution problem**, while DHCP is responsible for the IP-assignment problem.

Scenario 6.

❖ After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

❖ Question: Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

➤ **Firewall Troubleshooting**
Troubleshooting Steps:

- Check connectivity: Try to reach the internal application server by its IP address and verify that the server itself is running.
- Check firewall rules: Review the firewall rules for the server's IP address, port, and protocol.
- Check firewall logs: Look for blocked or denied connection attempts from employees' devices.
- Test the required port: Use a suitable connectivity test to confirm whether the application server's required port is being blocked.
- Correct the rule: If legitimate traffic is being blocked, create or modify a firewall rule to allow the required traffic while keeping other unnecessary traffic blocked.
- Retest: Ask an employee to access the application again and confirm that it works correctly.

Resolution:

If the firewall is blocking legitimate application traffic, **correct the firewall rule for the required IP, port, and protocol**. Then retest the connection to confirm the internal application server is accessible.

❌ Section B – Practical Application

Task 1.

❖ Workstation Assembly.



Figure 1: Workstation Assembly - Component Installation Sequence and POST Verification.

🔧 Component Installation

Installed the **CPU**, **CPU cooler**, **RAM**, **SSD**, **power supply**, and **graphics card** into the chassis in the correct sequence.

🔗 **Cable Connections**

Connected all required **power, data, CPU fan, GPU power, and front-panel cables** properly.

🔗 **Installation Documentation**

Documented the complete installation process **step-by-step from preparing the chassis to closing the case.**

🔗 **POST Verification**

Powered on the workstation and confirmed that it **successfully passed POST (Power-On Self-Test)** with the hardware detected correctly.

Result:

The workstation was **assembled successfully and passed POST**, confirming that the major components and connections were installed correctly.

❖ RAM Mismatch Troubleshooting.

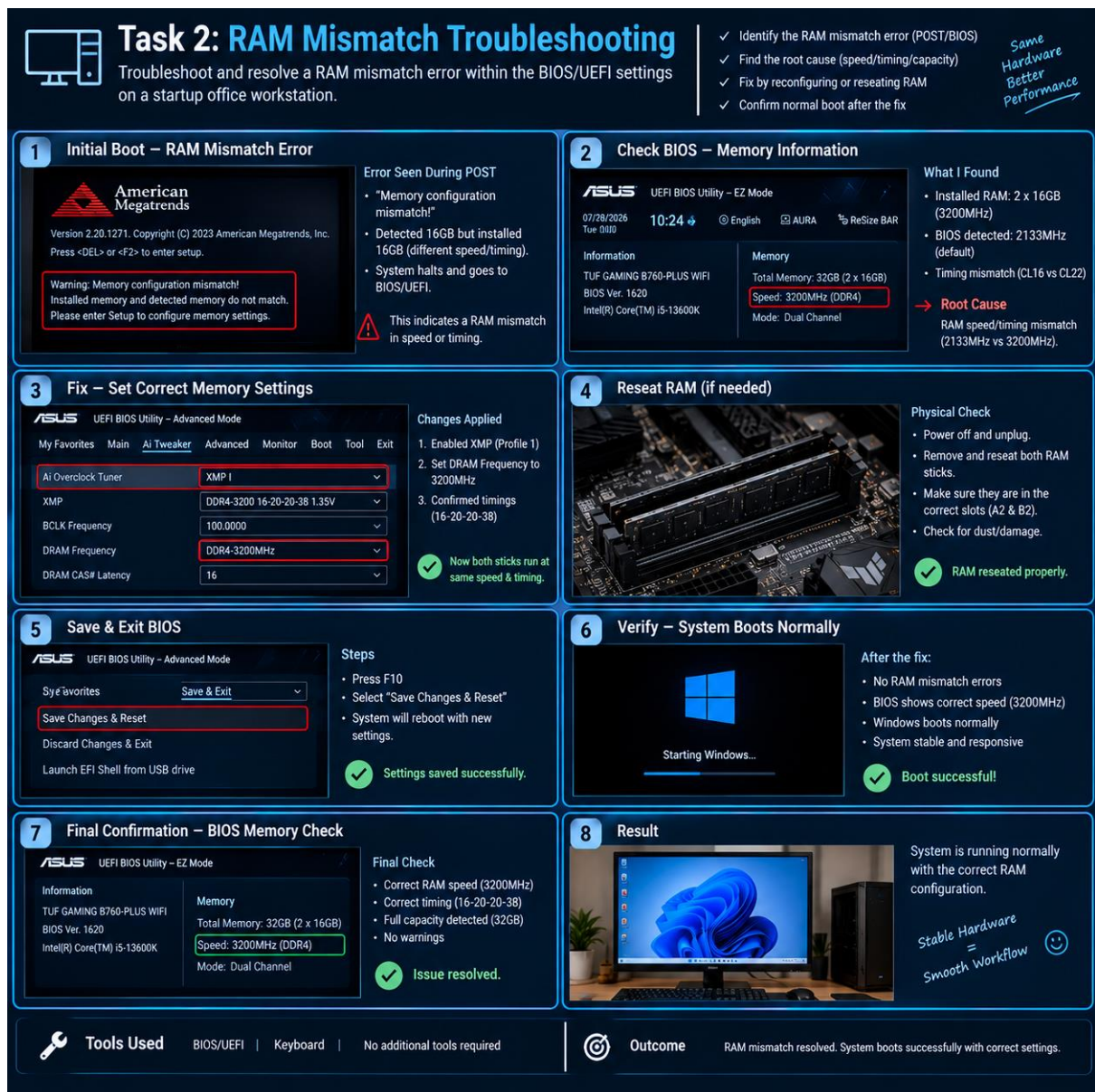


Figure 2: RAM Mismatch Troubleshooting - Error Identification, Resolution and Boot Verification.

 Identify the Error:

During POST, the BIOS displayed a **“Memory configuration mismatch”** warning, indicating a RAM mismatch.

➤ **Determine the Root Cause:**

The installed RAM had a **speed/timing mismatch**. The BIOS was detecting the RAM at 2133 MHz instead of the required 3200 MHz setting.

➤ **Resolve the Mismatch:**

The RAM was **reseated correctly**, and the BIOS/UEFI memory settings were reconfigured to use the correct **3200 MHz speed and matching timings**.

➤ **Confirm Normal Boot:**

After saving the BIOS settings and restarting, the system **booted normally without any RAM mismatch error**.

Result: The RAM mismatch was successfully resolved and the workstation became stable and ready for use.

❖ LAN Topology Build.

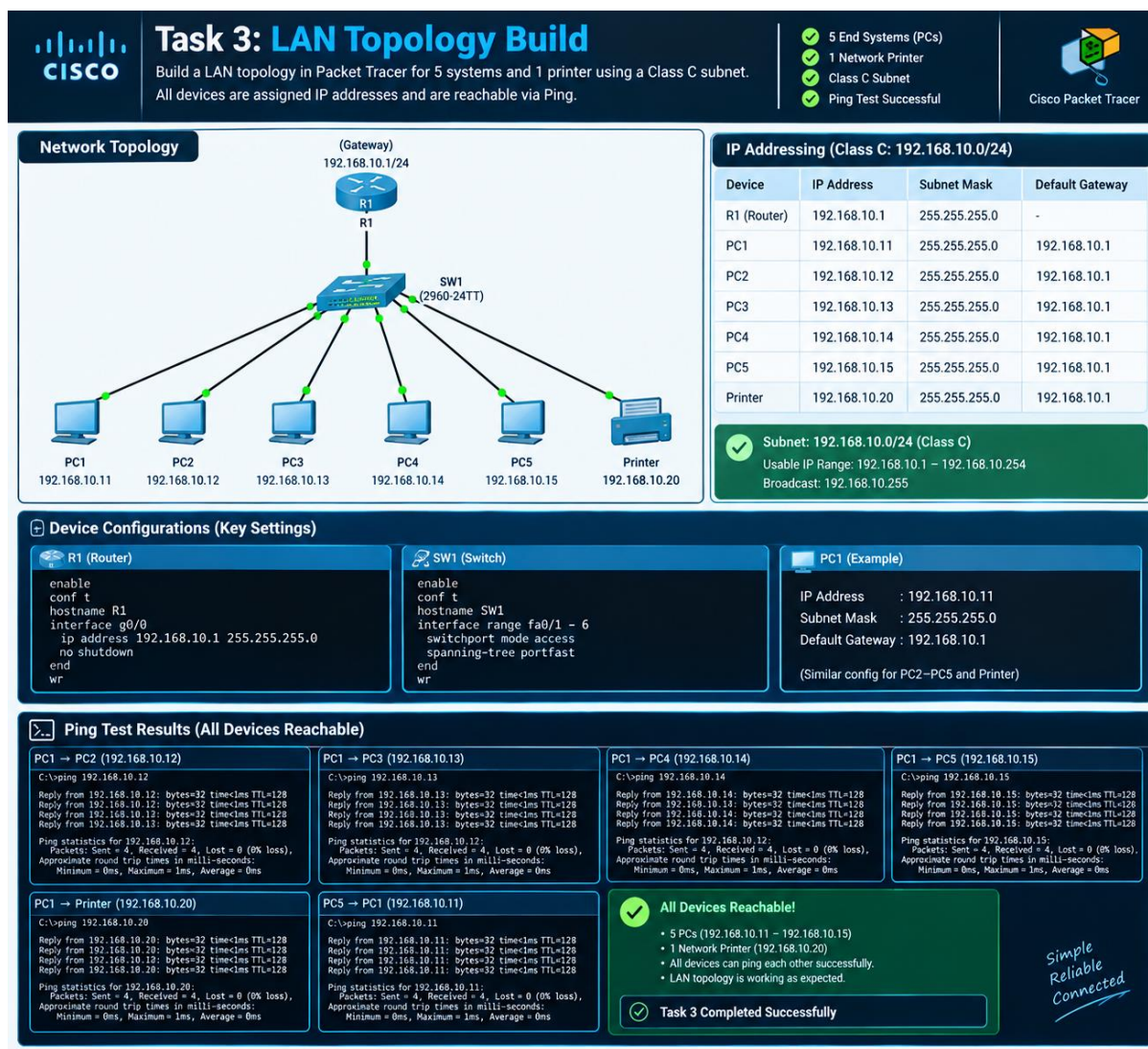


Figure 3: LAN Topology Build - IP Addressing and Ping Connectivity Verification.

Build the Topology:

The LAN topology was created in **Cisco Packet Tracer** using a switch to connect all devices.

➤ **Add Devices:**

The network includes **5 PCs and 1 network printer**, all connected to the switch.

➤ **Assign IP Addresses:**

All devices were assigned unique IP addresses from the **Class C subnet 192.168.10.0/24**.

➤ **Verify Connectivity:**

Ping tests were performed between the devices, and **all assigned IP addresses responded successfully**.

Result: The LAN topology was successfully built, configured, and verified with **100% successful connectivity**.

❖ Clean OS Installation.



Figure 4: Clean OS Installation - Manual Partitioning, Startup Optimization and Boot Verification.

Clean OS Installation:

Performed a clean installation of **Windows 11** using an **ISO installation media** on the workstation.

➤ **Manual Disk Partitioning:**

Created and configured the required **system, Windows, and data partitions manually** instead of using automatic partitioning.

➤ **Startup Optimization:**

Disabled **unnecessary startup applications and non-essential services** to improve the system's boot performance.

➤ **Boot Verification:**

Restarted the workstation and confirmed that **Windows 11 booted normally without any errors.**

Result: The OS was successfully installed, optimized, and verified for normal operation.

⌘ Section C – Mini Capstone Project

Task 1.

❖ Mini Project: Nexus Innovations - Diagnosing Random System Reboots.

 **Objective:** At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets, Analyze the probable hardware and thermal causes and propose a remediation plan, combining hardware, power, and troubleshooting concepts from this module.

Answer:

➤ 1. Three Hardware Components Most Likely to Cause Random Reboots:

- ⌘ **GPU (Graphics Card):** Generates high heat and draws significant power during rendering. Overheating can cause shutdowns or restarts.
- ⌘ **CPU (Processor):** Heavy workloads increase CPU temperature. Excessive heat can trigger thermal protection and restart the system.
- ⌘ **PSU (Power Supply Unit):** An overloaded, faulty, or unstable PSU may fail to provide consistent power during peak CPU/GPU loads.

Mini Project: Diagnosing Random System Reboots

At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets. Analyze the probable hardware and thermal causes and propose a remediation plan.

1 Three Most Likely Hardware Components

Under heavy load and poor ventilation, the following components are most likely to cause random reboots:

1. GPU (Graphics Card)

- High power draw during rendering
- Generates a lot of heat
- Can trigger thermal shutdown or instability

2. CPU (Processor)

- Runs at high temperatures under load
- Thermal throttling or overheating can cause unexpected restarts

3. PSU (Power Supply)

- Overheating inside cabinet
- Voltage drops under load
- May fail to provide stable power to the system

2 Why POST Passes in the Morning but Fails in the Afternoon?

Morning (Cooler Conditions)

- System is cool, components are at lower temperature
- Thermal headroom is available
- PSU delivers stable power

Afternoon (Heavy Load + Poor Ventilation)

- Rendering workload increases CPU/GPU usage
- Heat builds up inside the desk cabinet (poor airflow)
- Temperatures rise, components become unstable
- Can trigger thermal shutdown or power instability → random reboot

→ So, the systems pass POST in the morning because they start cool, but in the afternoon, heat and load cause instability, leading to reboots.

3 Using a Stress-Test Tool to Confirm the Issue

Tools: OCCT, FurMark (GPU), Prime95 (CPU), or AIDA64 (combined).

1 Run a combined stress test (CPU + GPU) for 30–60 minutes.

2 Monitor temperatures, clock speeds, and power usage using HWMonitor / HWINFO.

3 Watch for signs of thermal throttling, high temps (e.g., > 90°C), or power drops (e.g., Vcore/12V line instability).

4 If the system restarts, check logs (Event Viewer) for WHEA errors, kernel power 41, or thermal events.

If temps are very high or power drops before a reboot, it confirms a thermal or power issue.

4 Three Remediation Steps

1 Improve Cooling & Airflow

- Clean dust, ensure proper ventilation
- Add extra case/desk fans or move PCs to open space
- Keep cabinet doors open / use airflow grills

2 Stabilize Power Delivery

- Check PSU health and cables
- Use a higher quality PSU (80+ Bronze/Gold)
- Consider a UPS for voltage spikes/sags

3 Thermal & System Optimization

- Reapply thermal paste (if needed)
- Set proper fan curves / clean filters
- Update BIOS, GPU drivers, and monitor temps regularly

5 Is a PSU Upgrade Necessary?

Maybe Necessary (If...)

- PSU is old, low quality, or has been running for long hours.
- System shows power-related errors (e.g., kernel power 41, Event ID 63).
- Measured PSU output is unstable under load (voltage drops).
- Current PSU wattage is close to the total system draw (especially with GPU + CPU).

Maybe Not Necessary (If...)

- PSU is a good quality unit (80+ Bronze/Gold) and has enough wattage (e.g., 650W+ for high-end GPU).
- No power-related errors are found.
- The main issue is clearly thermal (high temps in cabinet).

Technical Justification

A PSU upgrade is only required if the existing PSU is underpowered, failing, or delivering unstable voltage under load. If the restarts are caused by overheating due to poor ventilation, improving cooling and airflow will solve the problem without needing a new PSU.

BETTER COOLING + STABLE POWER = NO MORE RANDOM REBOOTS

NEXUS INNOVATIONS

Figure 5: Nexus Innovations - Random System Reboot Diagnosis and Remediation Plan.

2. Why Systems Pass POST in the Morning but Fail in the Afternoon:

In the morning, the workstations are **cool**, so the CPU, GPU, and PSU operate within safe temperature and power limits. During afternoon rendering, continuous heavy load produces more heat. Since the computers are inside **poorly ventilated cabinets**, heat gets trapped and temperatures rise, which can cause thermal shutdowns or power instability and lead to random restarts.

3. Stress-Test to Confirm the Issue:

I would use a tool such as **OCCT** to stress the CPU and GPU together for about **30–60 minutes** while monitoring temperatures, power usage, and system stability. If temperatures become very high or power readings become unstable before a restart, it would indicate a **thermal or power-related problem**.

4. Three Remediation Steps:

- **Improve cooling and airflow:** Clean dust, improve cabinet ventilation, and add/upgrade case fans if required.
- **Stabilize power delivery:** Check the PSU, power cables, and connections; replace the PSU if it is faulty or insufficient.
- **Optimize thermal performance:** Replace old thermal paste, ensure proper fan operation, and keep BIOS/GPU drivers updated.

5. PSU Upgrade Justification

A **PSU upgrade is necessary** if the existing PSU is underpowered, faulty, or shows unstable voltage during heavy load. However, if the PSU has sufficient wattage, is good quality, and power remains stable during stress testing, an upgrade may **not be necessary**. In that case, the main cause is more likely to be **overheating due to poor ventilation**.

✂ Section D – Ai Augmented Learning

Task 1.

❖ Step 1 - Build with AI.

➤ 1. AI Generated IP Addressing Plan:

Item	Details
Subnet	192.168.10.0/28
Subnet Mask	255.255.255.240
Usable Host Range	192.168.10.1 – 192.168.10.14
Broadcast Address	192.168.10.15

2. IP Address Allocation:

Device	IP Address
Gateway	192.168.10.1
System 1	192.168.10.2
System 2	192.168.10.3
System 3	192.168.10.4
System 4	192.168.10.5
System 5	192.168.10.6
Printer	192.168.10.7

3. Addresses Reserved for Future Expansion:

The addresses **192.168.10.8 – 192.168.10.14** are reserved for future computers, printers, or other network devices.

4. Exact Prompt Used:

“Create an IP addressing and subnetting plan for Nexus Innovations' startup office network with 5 systems and 1 printer. Specify the subnet, subnet mask, usable host range, and broadcast address. Allocate a unique IP address to all 5 systems and the printer, and identify IP addresses that should be reserved for future network expansion. Keep the plan simple and easy to understand.”

❖ Step 2 - Test & Debug (Without AI).

➤ 1. Exact Prompt Used:

“Create a simple IP addressing and subnetting plan for Nexus Innovations' startup office network covering 5 systems and 1 printer. Specify the subnet, subnet mask, usable host range, and broadcast address. Allocate a unique IP address to all 5 systems and the printer, and identify which IP addresses should be reserved for future expansion.”

2. AI's Original Plan and Corrected Version:

Device	AI's Original IP	Corrected IP
System 1	192.168.10.1	192.168.10.1
System 2	192.168.10.2	192.168.10.2
System 3	192.168.10.3	192.168.10.3
System 4	192.168.10.4	192.168.10.4

System 5	192.168.10.5	192.168.10.5
Printer	192.168.10.6	192.168.10.6
Future Expansion	192.168.10.7–.14	192.168.10.7–.14

⌘ **Subnet:** 192.168.10.0/28

⌘ **Subnet Mask:** 255.255.255.240

⌘ **Usable Host Range:** 192.168.10.1–192.168.10.14

⌘ **Broadcast Address:** 192.168.10.15

3. Verification Note:

I checked the subnet, host range, and broadcast address manually. The AI plan was **technically correct**, with enough host addresses for 5 systems, 1 printer, and future expansion. No overlapping or incorrect addresses were found, so **no changes were required**.